



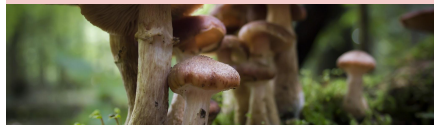
Mushroom Edibility Classification

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Background



Datasets



Methods



Results



Background

- Misclassification of mushrooms can be detrimental to health if intended for consumption
- The UCI Primary Mushroom Dataset (Schlimmer 1987) is a very widely used dataset for computational mushroom edibility classification
 - It has 8,124 hypothetical samples from 23 species
 - Linearly separable, and very small
- UCI Secondary Mushroom Dataset developed to be more real-world representative (Wagner 2021)
 - 173 species represented with 61,069 samples
- Danish Fungi 2020(DF20) dataset
 - 295,938 images across 1,604 species
- Picek et al. demonstrates good species identification performance using CNN architectures on the DF20 dataset
- Grinsztajn et al. provide an analysis demonstrating that tree-based models tend to outperform deep learning methods on medium-sized tabular data.

Project Goal

How does CNN image classification compare to tabular ML classification when the tabular dataset is restricted to visually observable features?

Project Pipeline:

Secondary
Mushroom
Dataset

Preprocessing
Dropped cols |
OHE

Model Training:
LR | SVM | kNN | RF
| XGBoost | MLP

5-fold
stratified CV

Results

Comparison

DF20 Image
Dataset

Preprocessing
Edibility labels

Preprocessing
Stratified Split

ResNet18
fine-tuning

Results

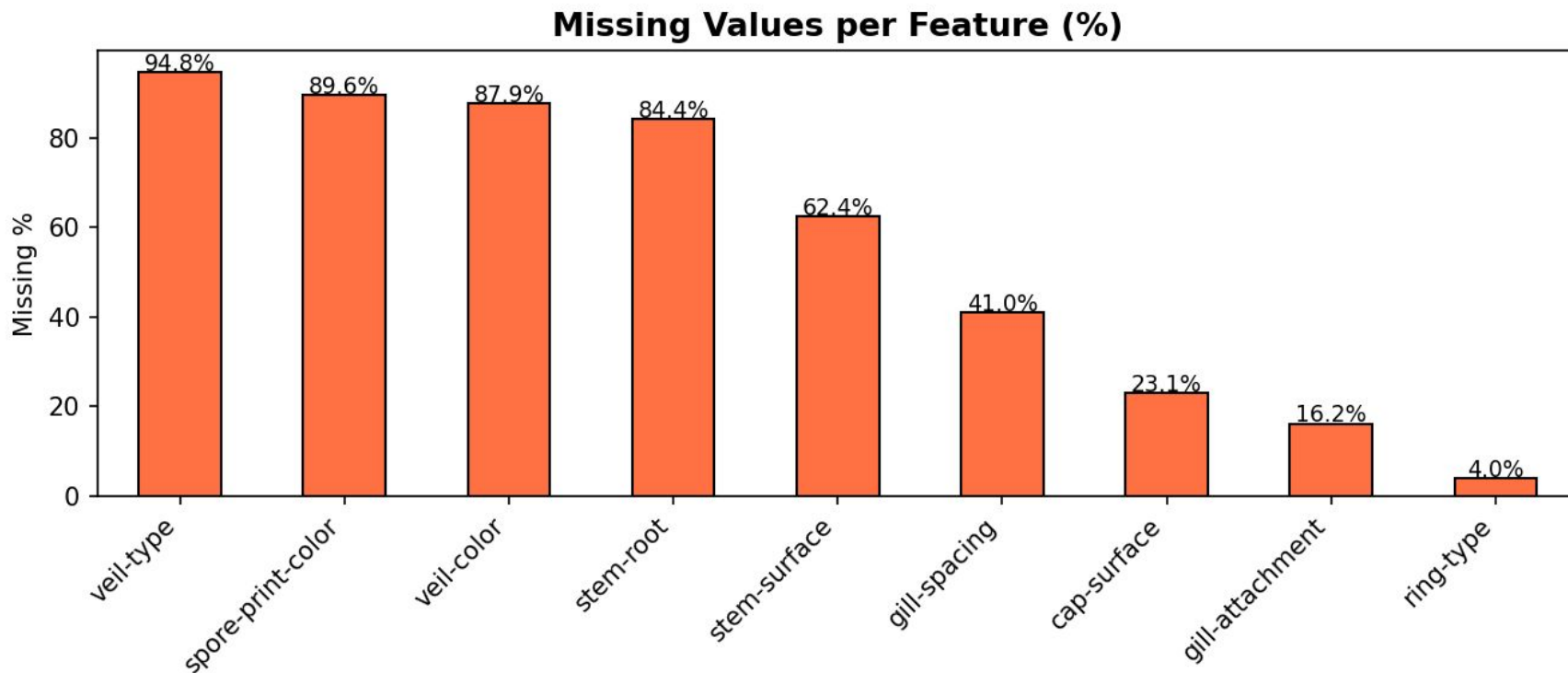
Tabular Dataset

- Secondary Mushroom Dataset from UCI Machine Learning Repository(2021)
 - Synthetically generated dataset based off of species descriptions in book *Mushrooms and Toadstools* for mushrooms commonly found in Europe(Wagner 2021)
- 61,069 samples across 173 species
 - Only stemmed mushrooms included
- Each sample labeled edible or poisonous (44.5% to 55.5% ratio)
- 20 features (3 continuous, 17 categorical)

Tabular Dataset - Preprocessing

- Drop columns that would not be visible in a photograph
- Drop columns with more than half missing values or zero variance
 - Final Features used:
 - Continuous: ['cap-diameter', 'stem-height', 'stem-width']
 - Categorical: ['cap-shape', 'cap-surface', 'cap-color', 'gill-attachment', 'gill-spacing', 'gill-color', 'stem-color', 'has-ring', 'ring-type']
- Mean(continuous features) mode(categorical) imputation for columns with some missing values
- StandardScaler on continuous features
- OneHotEncoder on categorical features
 - 9 features to 75 features after OHE
- Clip values to [-10, 10] to prevent overflow from continuous features in LR
- Stratified 70:15:15 train:val:test split

Tabular Dataset - Preprocessing



Methods - Tabular Dataset Models

Baseline

- Logistic Regression

Classical

- SVM
- kNN

Ensemble

- Random Forest
- XGBoost

Deep Learning

- MLP (MultiLayerPerceptron)

Performance Metrics:

- Accuracy, Precision, Recall, F1 score, AUC-ROC

Results - Tabular Classification

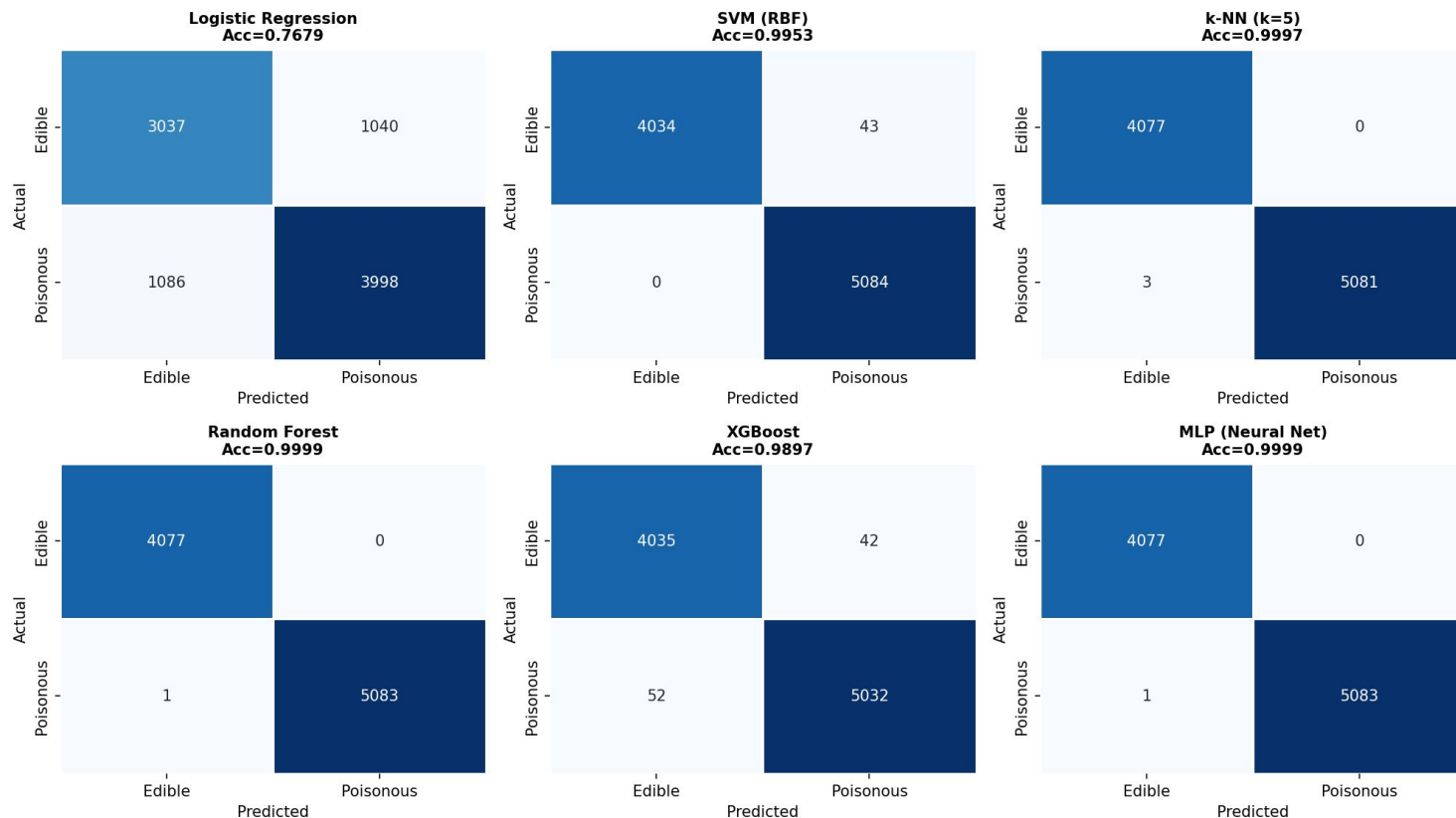
5-fold stratified cross-validation results on the training set (tabular models only). MLP was only evaluated on test set.

Model	Accuracy	$\pm$ Std	F1	AUC
Random Forest	0.9999	0.0001	0.9999	1.0000
k-NN ($k=5$)	0.9997	0.0001	0.9997	1.0000
SVM (RBF)	0.9922	0.0015	0.9930	0.9986
XGBoost	0.9814	0.0011	0.9832	0.9982
MLP (PyTorch)	—	—	—	—
Logistic Regression	0.7679	0.0057	0.7890	0.8364

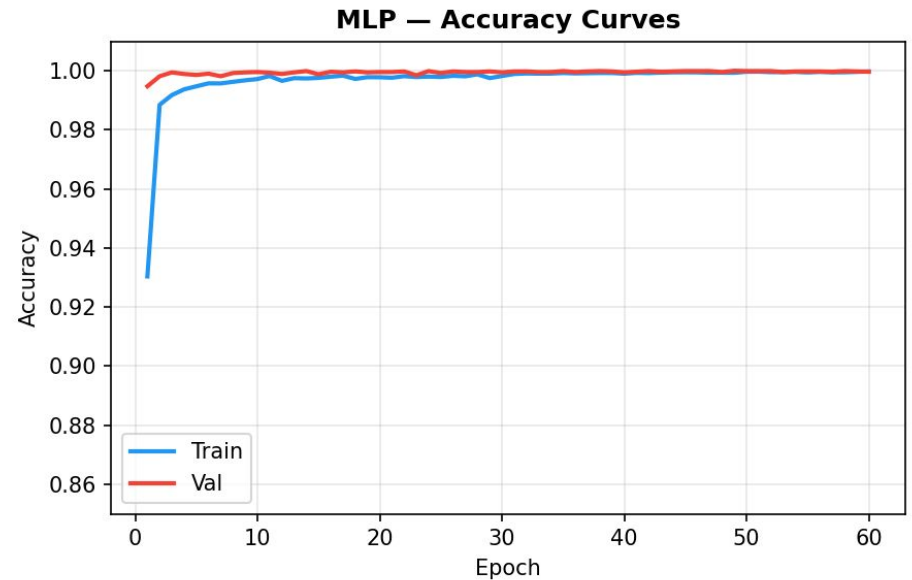
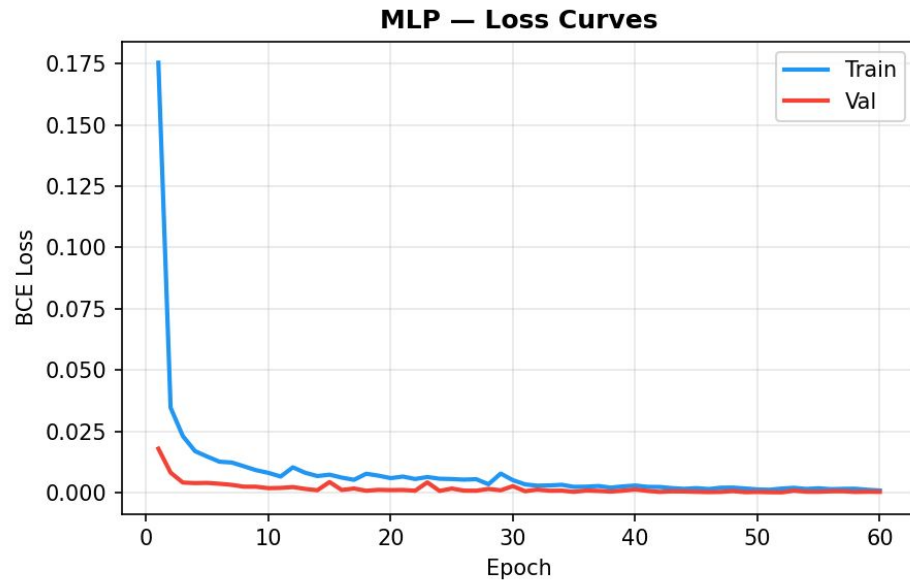
Test Set Results For All Models.
Note the test set for CNN is different because it is image-based

Model	Type	Acc.	F1	AUC	Prec.	Recall
Random Forest	Ensemble	0.9999	0.9999	1.0000	1.0000	0.9998
MLP (Neural Net)	Deep Learn.	0.9999	0.9999	1.0000	1.0000	0.9998
k-NN ($k=5$)	Classical	0.9997	0.9997	1.0000	1.0000	0.9994
SVM (RBF)	Classical	0.9953	0.9958	0.9995	0.9916	1.0000
XGBoost	Ensemble	0.9897	0.9907	0.9991	0.9917	0.9898
Logistic Regression	Baseline	0.7679	0.7900	0.8361	0.7936	0.7864
ResNet18 (CNN)	Image	0.8337	0.8408	0.9188	0.8341	0.8476

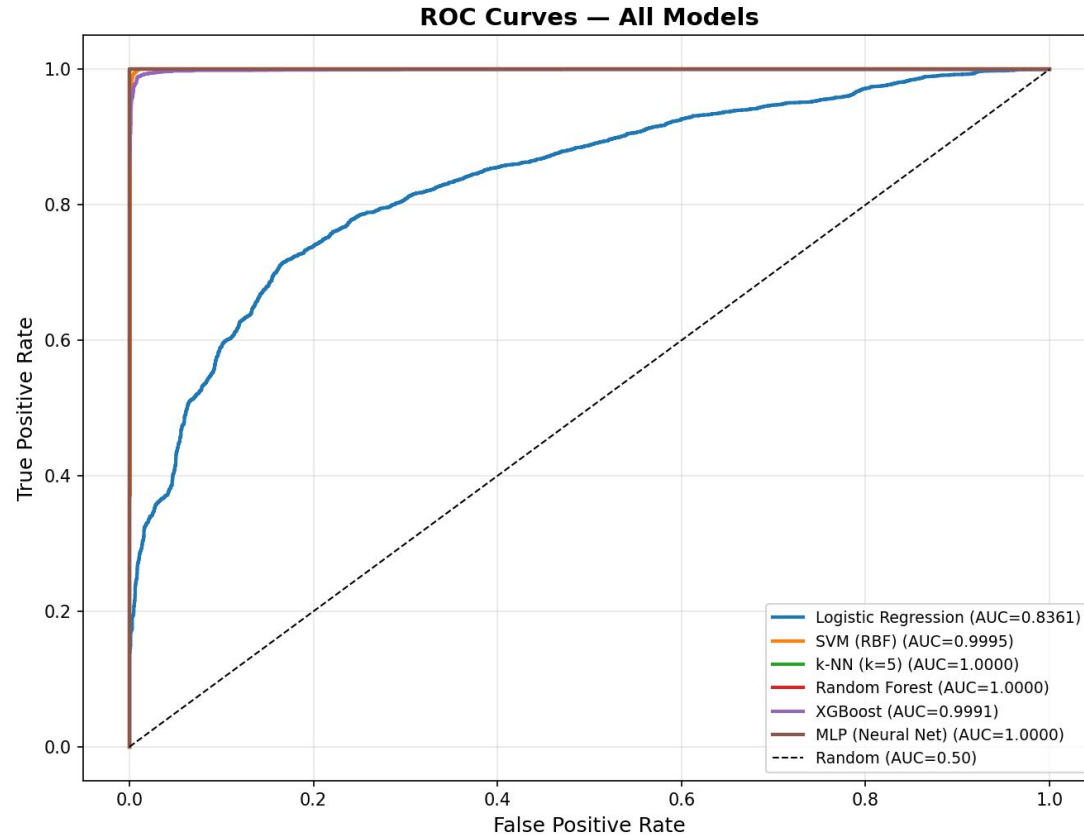
Confusion Matrices — Test Set



Confusion matrices for all tabular models on the held-out test set. Non-linear classifiers (Random Forest, MLP, k-NN) produce near-zero false negatives (poisonous predicted as edible), while Logistic Regression shows substantially higher error rates on both classes.



MLP Training Curves - converged within 5 epochs and plateaued near 100% val accuracy



ROC curves for all tabular models on the held-out test set. Random Forest, MLP, and k-NN achieve AUC of 1.000. Logistic Regression achieves AUC of 0.836, confirming the non-linear structure of the classification problem.

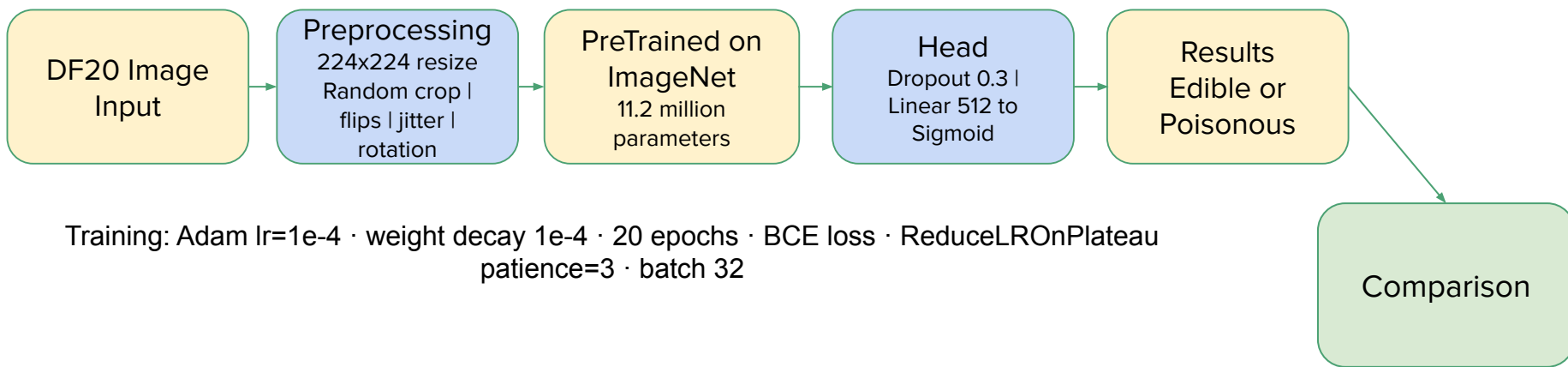
Image Dataset

- Danish Fungi 2020
- 45 overlapping species with tabular dataset
 - Each species having a threshold of at least 50 images available
- 21,580 images for those species
- In the original DF20, each image is mapped to a ground truth species label
 - I then mapped each species to an edible/poisonous label
- Out of the 21,580 images, each species was proportionally divided into 70:15:15 train:val:test split

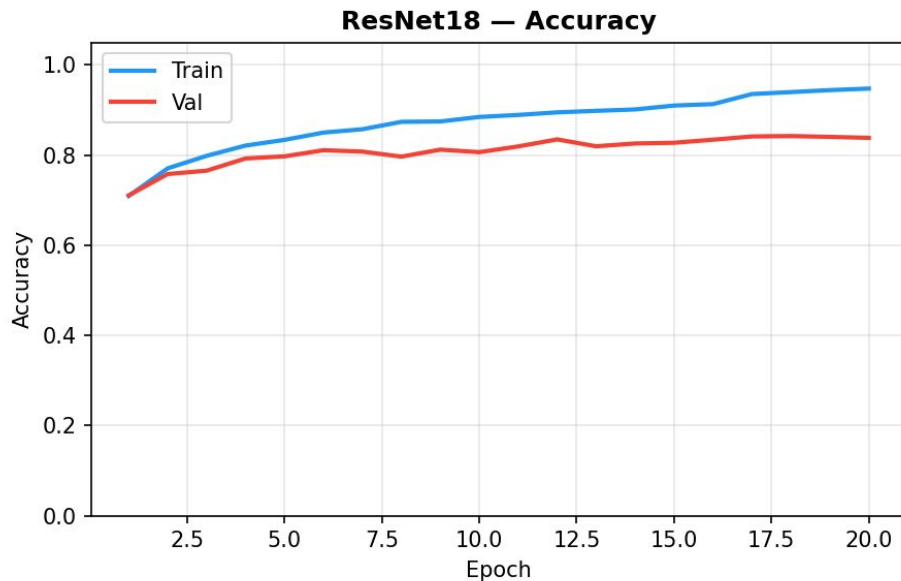
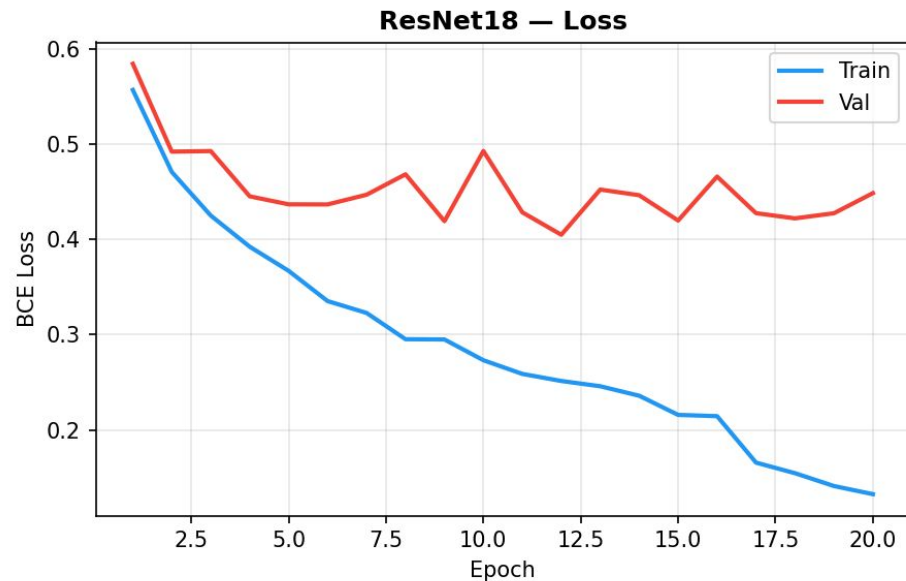


Actual reference images from DF2020

Methods - Image Dataset Model

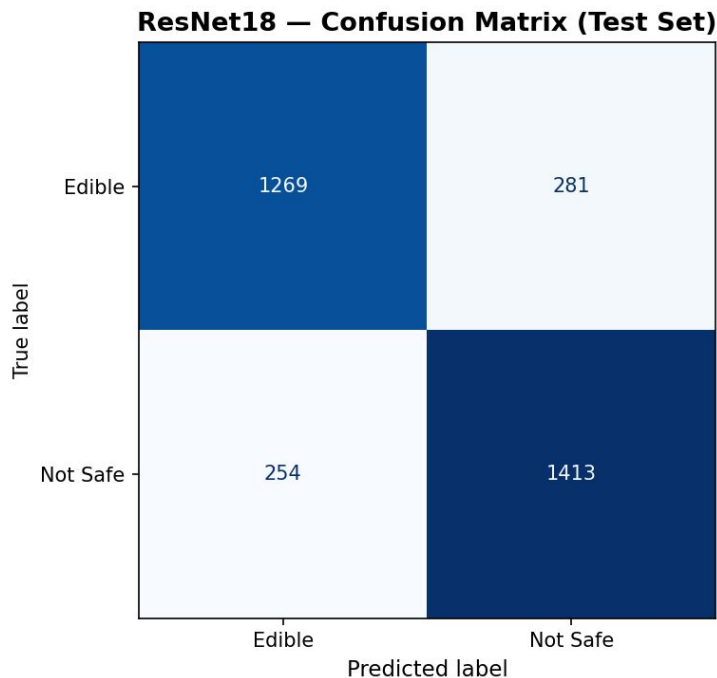


Results

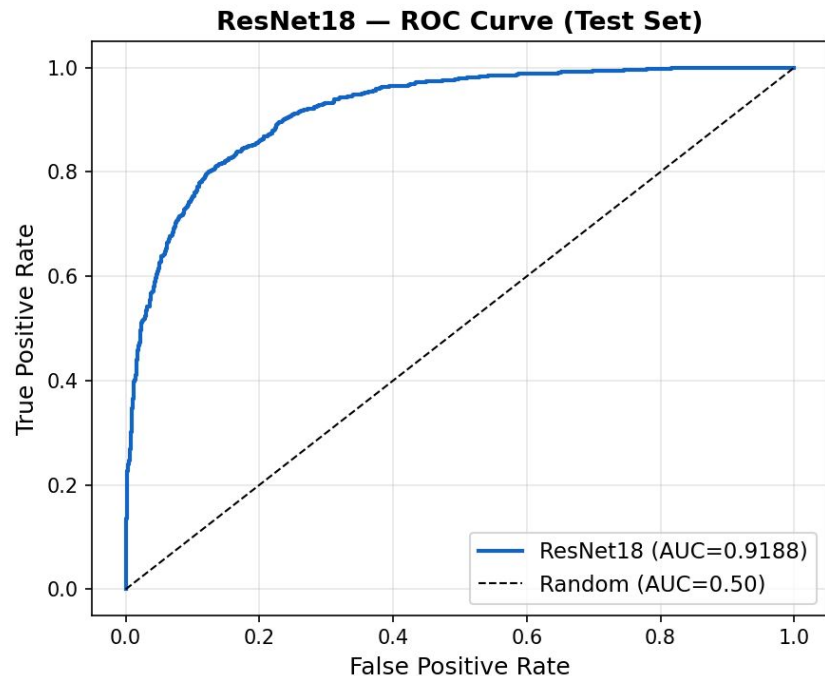


ResNet18 training and validation loss (left) and accuracy (right) over 20 epochs. The learning rate scheduler triggers at epoch 16, producing a visible improvement in both curves. The best checkpoint is saved at epoch 18 (minimum validation loss).

Results



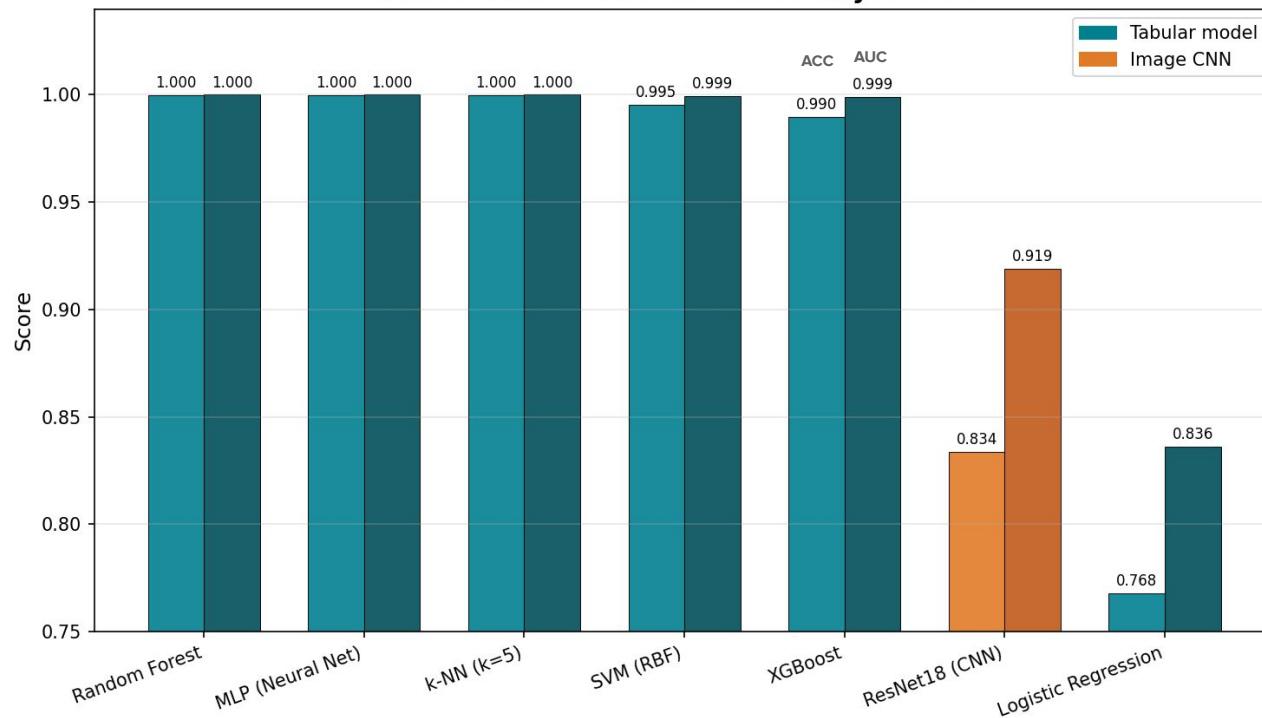
ResNet18 confusion matrix on the test set. 254 poisonous samples are misclassified as edible (false negatives), representing 15.2% of all poisonous test samples. False positives number 236.



ResNet18 ROC curve on the test set (AUC = 0.9188). The high AUC relative to the 83.4% accuracy indicates that the model has learned discriminative visual features, with the accuracy ceiling potentially imposed by visually ambiguous look-alike species.

Comparison Results

Tabular Models vs CNN — Accuracy and AUC



Darker bar is the AUC score

Lighter is the test accuracy.

Discussion

Tabular

- For this dataset, even with the models reduced to only visible features most models did well
- LR failed because dataset was not linearly separable

Review of Features

- Continuous: ['cap-diameter', 'stem-height', 'stem-width']
- Categorical: ['cap-shape', 'cap-surface', 'cap-color', 'gill-attachment', 'gill-spacing', 'gill-color', 'stem-color', 'has-ring', 'ring-type']

Image

- The CNN plateaued in Test Accuracy around 15 epochs, saved epoch 18 with lowest validation loss
- Although the image classification did well, perhaps more precise measurements in the tabular categories allowed for more accurate classification
 - I.e. look-alike mushrooms may have precise feature distinctions that are not detectable in photograph

Limitations

- Image data and tabular datasets are different(Secondary UCI vs DF20)
- Results would be more meaningful if the datasets were the same,
 - I.e. if each instance was collected as an image, and then measurements were made to collect the tabular data
- More species represented in tabular dataset with more training points
 - Overlap of 45 species
- Partially observable features are fully represented in the tabular dataset
 - Especially the continuous features, these need something present to scale the objects in the image

References

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Thank you!
